# Nethmi Jayasinghe

**Electrical and Computer Engineering, PhD Student**

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## Profile

PhD student in Electrical and Computer Engineering working on reinforcement learning and robust, adaptive AI for robotics and embodied systems. My research develops learning algorithms that let agents adapt *online*–recovering from dynamics shifts and post-training faults at inference time without retraining–by combining deep RL, control, and brain-inspired learning, validated in PyTorch across MuJoCo locomotion benchmarks and real robotic platforms (quadruped, biped, humanoid, wheeled). First-author work accepted at **ICML 2026** and under review at **CoRL 2026** and **AAAI 2027**.

## Research Interests

Reinforcement Learning · Robot Learning & Embodied AI · Robust & Adaptive ML · Deep Learning · Representation & Multi-Modal Learning

## Skills

- **Machine Learning & RL:** PyTorch, Deep Reinforcement Learning (SAC, PPO, DQN, Actor–Critic), Online / Inference-Time Adaptation, Deep Neural Networks, Transformers, Representation Learning, scikit-learn, TensorFlow
- **Robotics & Simulation:** MuJoCo, Gymnasium, Webots, Sim-to-Real Transfer, Real-Robot Deployment (ViperX robot arm, Agilex Scout Mini Pro), Manipulation & Locomotion Control
- **Programming & Engineering:** Python, SQL, MATLAB, Linux, Git, Large-Scale Experimentation Pipelines

## Education

**University of Illinois Chicago** **USA** *2023-Present*
Ph.D. in Electrical and Computer Engineering
Expected Graduation: May 2028
GPA: 3.9/4.0

**University of Moratuwa** **Sri Lanka** *2017-2022*
B.Sc. (Hons) in Electrical Engineering
GPA: 3.81/4.2

## Research Experience

**Graduate Research Assistant – AEON Lab, University of Illinois Chicago** *Advised by Prof. Amit Trivedi* *2023 - Present*

- Lead research on reinforcement learning, continual robot learning, and inference-time adaptation–producing first-author work accepted at ICML 2026 and under review at CoRL 2026 and AAAI 2027 (see Publications)–running large-scale PyTorch experiments across MuJoCo benchmarks and real robotic hardware (ViperX arm, Agilex Scout Mini Pro).

## Publications & Preprints

**Cerebellar-Inspired Residual Control for Fault Recovery: From Inference-Time Adaptation to Structural Consolidation**
*Accepted, ICML 2026*

- Developed a residual-control architecture that augments a frozen RL policy with a parallel online correction pathway, enabling inference-time adaptation to post-training faults without retraining, via modular error-driven, dual-timescale plasticity with performance-driven gating.
- Evaluated on MuJoCo locomotion (HalfCheetah-v5, Ant-v5, Humanoid-v5) and PandaReach-v3, achieving up to **+66%** on HalfCheetah-v5 and **+53%** on Humanoid-v5 without policy retraining.

**Fast and Efficient Emergent Differentiation for Novelty Detection Driven by Cerebellum-Inspired Memtransistors**
*Accepted, Nature Communications, 2026*

- Built a real-time anomaly/novelty-detection system combining transformer-based temporal encoding with spiking neural networks for ECG arrhythmia detection within a single heartbeat, using $>\mathbf{10^4}\times$ **fewer operations** than Transformer baselines.
- Generalized the framework to multi-modal inputs (ECG, images, audio) with robustness via ensemble neuron consensus.

**When Robots Sleep: Offline Skill Consolidation for Shared-Policy Robot Learning**
*Under review, CoRL 2026*

- Proposed *Sleeping Robots*, a wake–sleep framework for continual robot skill learning that consolidates a single shared policy offline using compact frozen skill memories (frozen critics for RL, actor snapshots for imitation), without trajectory replay or task-specific heads, routing, or adapters.
- On Meta-World MT5, improved average success by **+64%** and pairwise reliability by **2.0**× over the strongest non-oracle baseline; on SurgicAI, achieved the best average success and backward transfer among continual baselines.
- Validated the consolidated policy on a physical ViperX 6-DOF robot arm, demonstrating real-robot skill retention beyond simulation.

**Stability-Aligned Residual Adaptation for Rapid Recovery from Dynamics Shifts**
*Under review, AAAI 2027*

- Formulated rapid inference-time recovery as constrained residual learning and introduced a Stability Alignment Gate (magnitude bounds, directional coherence, performance-conditioned gain) that regulates an online residual's corrective authority around a frozen RL policy–no retraining, resets, or system identification.
- Cut recovery time by up to **87%** (Go1), **48%** (Cassie), **30%** (H1), and **20%** (Scout) vs. a frozen SAC baseline across four morphologies, with real-robot validation on a ViperX arm and Agilex Scout Mini (**19.5%** on hardware).

**GaitVista: Reliability-Aware AI Measurement toward Accessible Longitudinal Gait Assessment**
*Under review, AAAI 2027*

- Built a reliability-aware visual–inertial fusion layer for accessible 3D gait assessment: a lightweight gate assigns per-joint, per-frame modality contributions from sensing quality, cross-modal disagreement, and motion continuity, without condition labels or error supervision.
- Reduced average lower-body error by ∼**27.8%** across seven degraded sensing conditions on TotalCapture and, on MoVi, was the only deployable fusion method to improve over both unimodal streams.

**Single-Step Extraction of Transformer Attention with Dual-Gated Memtransistor Crossbars**
*IEEE Electron Device Letters, 2024*

- Developed a hardware–software co-design framework that accelerates Transformer attention by extracting attention scores directly, eliminating explicit query/key computation.
- Reduced operations and memory overhead, achieving **2.37**× **lower energy** on multiple time-series datasets.

## Industry Experience

**Research and Development Engineer – Synopsys Inc.** *Sri Lanka* *06/2022 - 07/2023*

- Designed and deployed a SQL-based database system for large-scale test-case data, reducing retrieval latency and improving analysis efficiency.
- Built an interactive Python/SQL analytics dashboard for real-time visualization of validation metrics, and automated failure-pattern detection and root-cause analysis in Python, accelerating debugging.

## Teaching

**Graduate Teaching Assistant** *University of Illinois Chicago* *2023 - Present*
ECE 115: Introduction to Electrical and Computer Engineering
ECE 350: Principles of Automatic Control